Security Requirements

Req ID	Description	Priority	Acceptance Criteria	Rationale (short)
SR-1	The system shall validate the selected input file before beginning compression or decompression.	High	When a valid and accessible file is selected, the system accepts it. If the file does not exist or cannot be opened, the system displays an appropriate error message.	Prevents invalid file access and processing errors.
SR-2	The system shall validate compressed files before decompressing them to ensure that the input follows the expected file format.	High	Valid compressed files are processed. Files with invalid headers, malformed data, or unexpected formats are rejected safely.	Prevents malformed input from disrupting decompression.
SR-3	The system shall prevent accidental overwriting of existing files during compression and decompression.	High	If the specified output file already exists, the system requests confirmation or rejects the operation without modifying the existing file.	Protects existing user data from accidental loss.
SR-4	The system shall preserve the integrity of the original file during compression.	High	After compression and subsequent decompression, the recovered file matches the original file byte-for-byte for all supported inputs.	Ensures lossless and reliable data recovery.
SR-5	The system shall handle empty files, corrupted input, and unexpected end-of-file conditions without crashing.	High	Empty, corrupted, or incomplete input files are handled gracefully, with appropriate error messages and no abnormal termination.	Improves robustness against invalid input.
SR-6	The system shall validate file sizes and input data to reduce excessive memory consumption during compression and decompression.	Medium	Inputs exceeding configured processing limits are handled gracefully without crashing or corrupting existing files.	Reduces the risk of resource exhaustion.
SR-7	The system shall restrict file operations to the input and output paths specified by the user and shall not modify unrelated files.	High	The system reads the specified input and writes to the intended output location without modifying unrelated files.	Prevents unintended file modifications.
SR-8	The system shall display appropriate error messages without exposing unnecessary internal implementation details.	Medium	File operation failures produce meaningful error messages without exposing memory contents, internal data structures, or sensitive system information.	Reduces unnecessary information disclosure.
SR-9	The system shall not transmit user file contents to external services unless explicitly required and authorized.	High	When operating as a standalone application, file processing occurs locally, and no file contents are transmitted to external services.	Protects user file confidentiality.
SR-10	The system shall safely handle file-processing errors and failed write operations.	High	If the system cannot write the output, it reports the failure and does not present an incomplete file as a successful result.	Prevents misleading results and incomplete files.